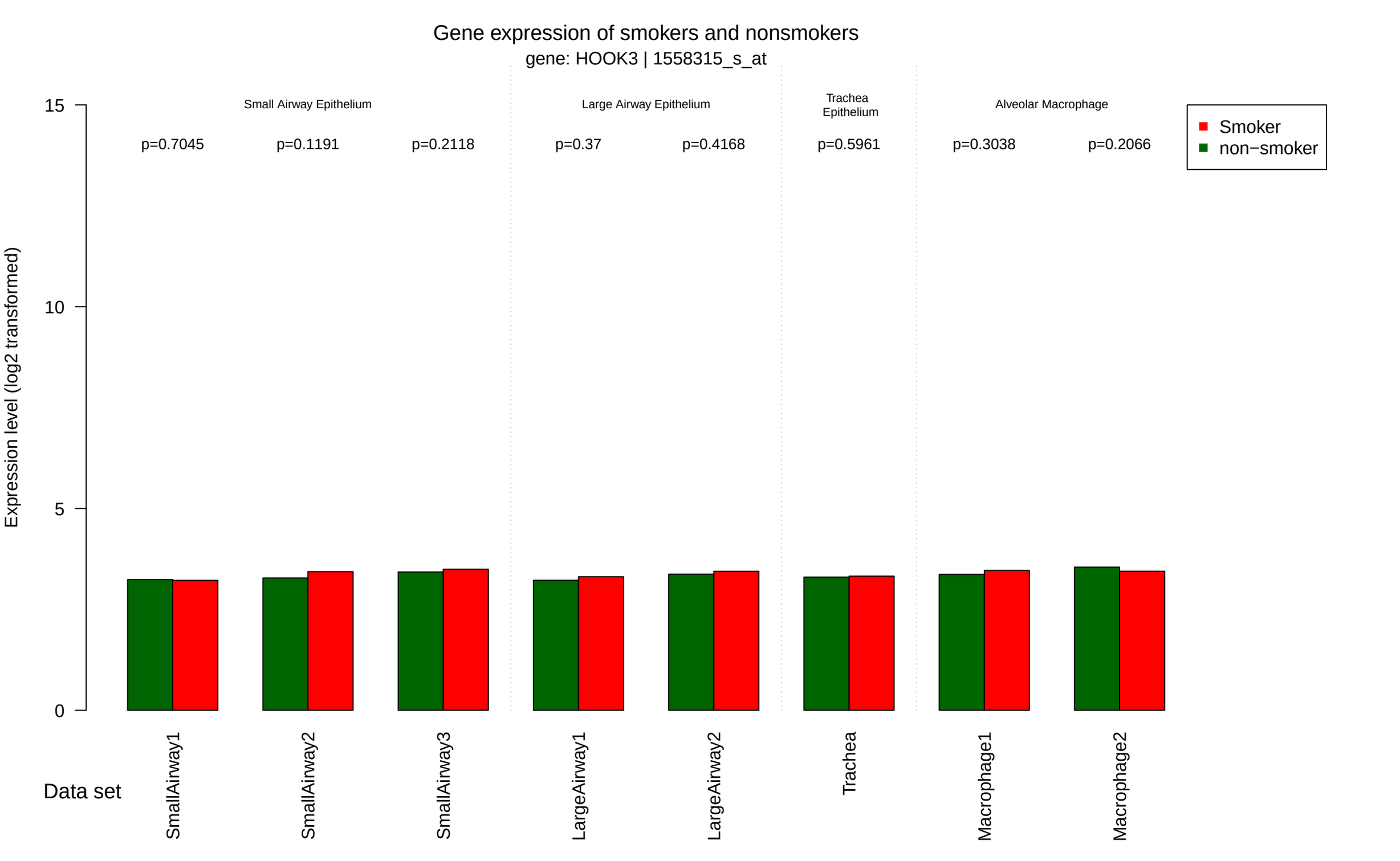
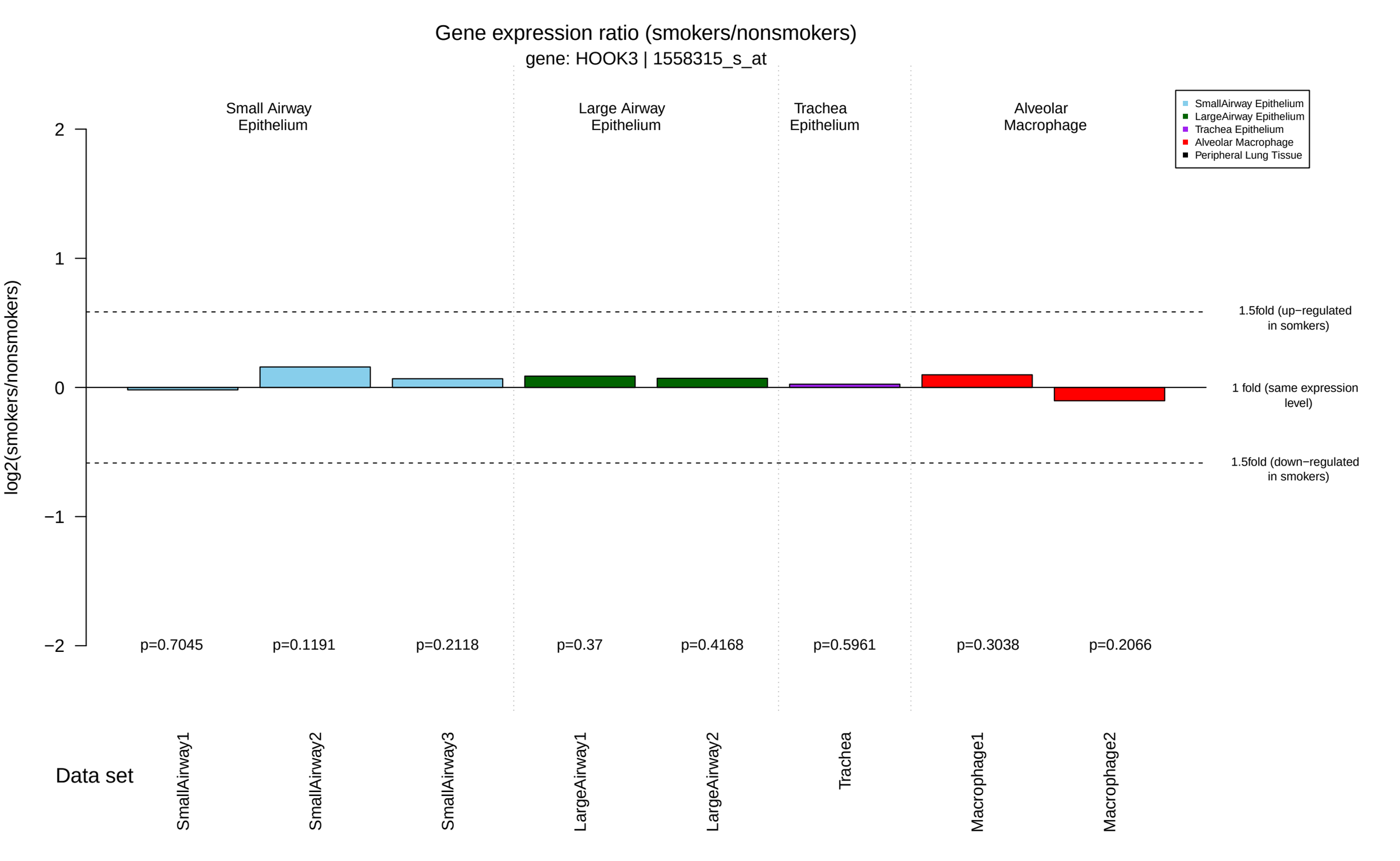
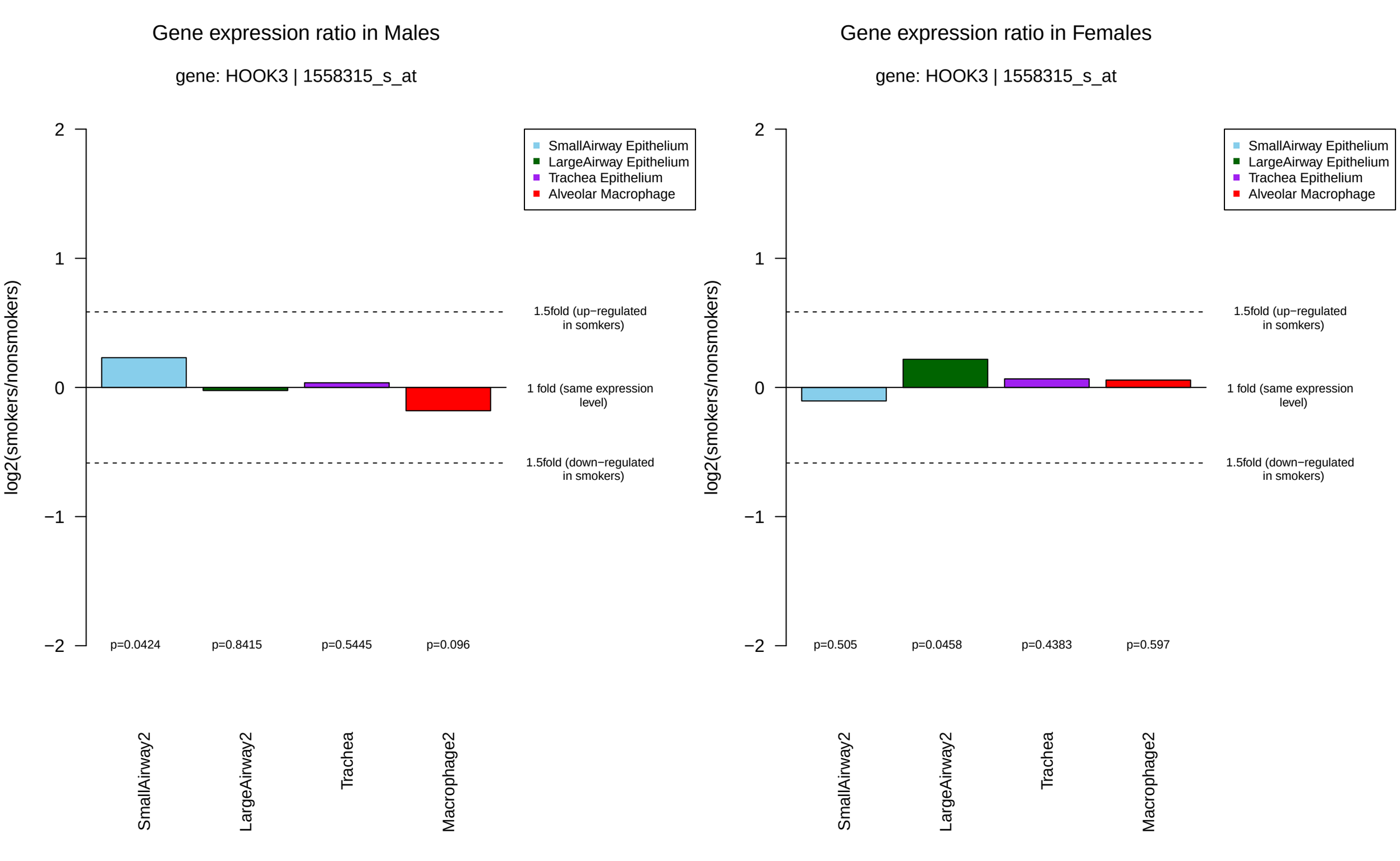
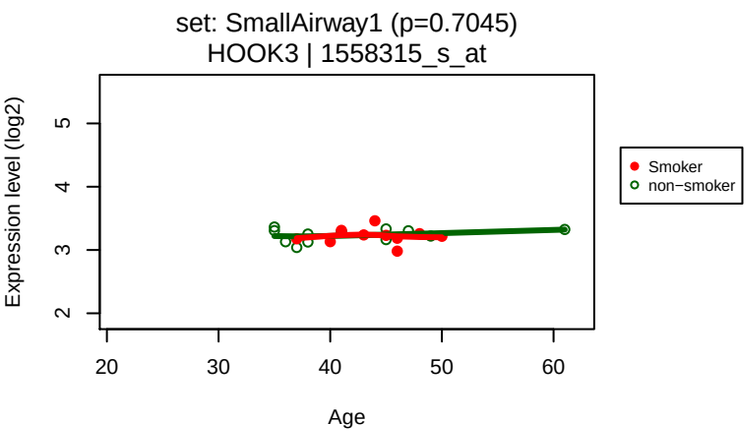
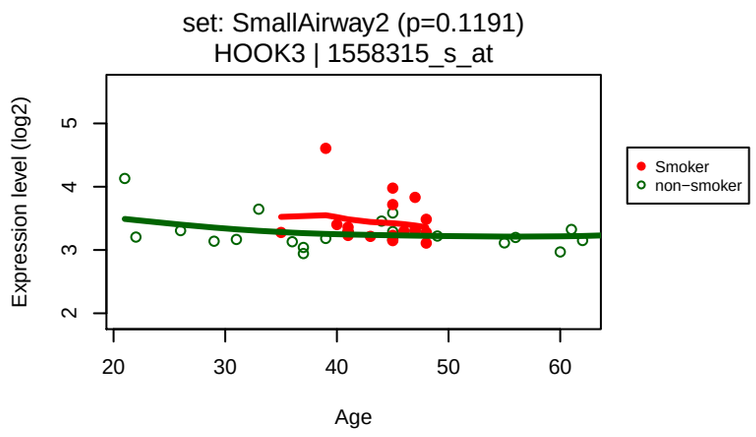
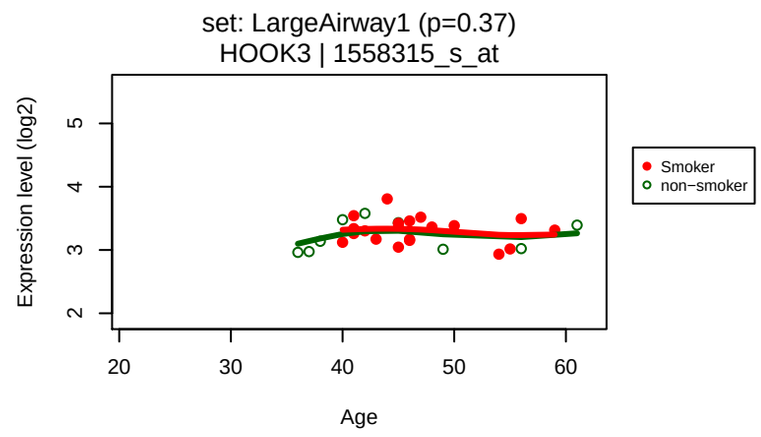

Gene expression of smokers and nonsmokers
gene: HOOK3 | 1558315_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7045
p=0.1191
p=0.2118
p=0.37
p=0.4168
p=0.5961
p=0.3038
p=0.2066
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

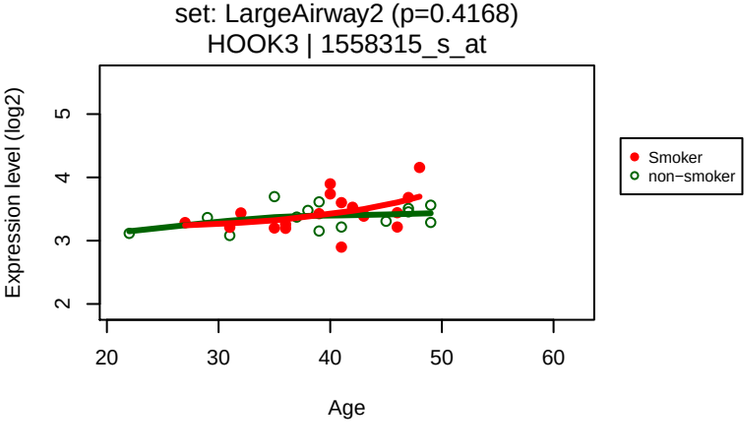

Gene expression ratio (smokers/nonsmokers)
gene: HOOK3 | 1558315_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7045
p=0.1191
p=0.2118
p=0.37
p=0.4168
p=0.5961
p=0.3038
p=0.2066
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

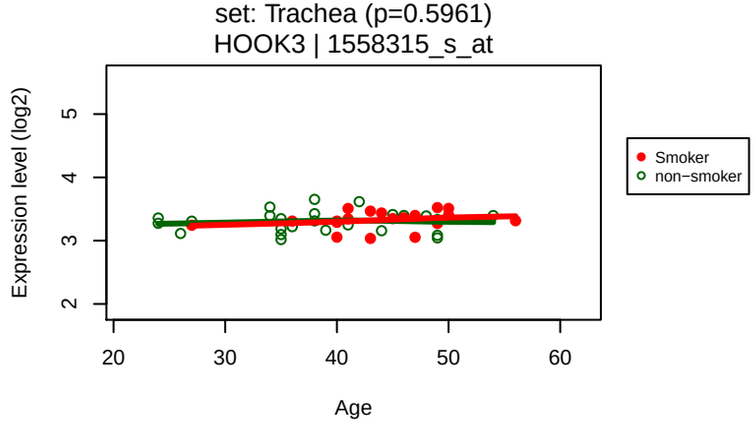

Gene expression ratio in Males
gene: HOOK3 | 1558315_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0424
p=0.8415
p=0.5445
p=0.096
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOOK3 | 1558315_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.505
p=0.0458
p=0.4383
p=0.597
SmallAirway2
LargeAirway2
Trachea
Macrophage2

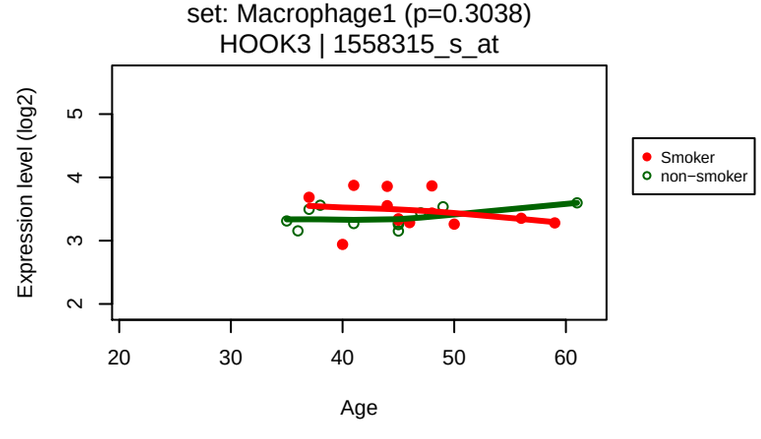

set: SmallAirway1 (p=0.7045)
HOOK3 | 1558315_s_at
Expression level (log2)
Age
Smoker
non−smoker

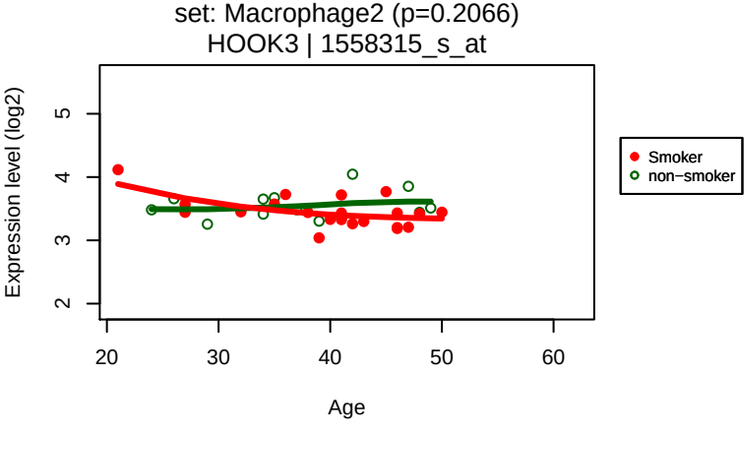

set: SmallAirway2 (p=0.1191)
HOOK3 | 1558315_s_at
Expression level (log2)
Age
Smoker
non−smoker

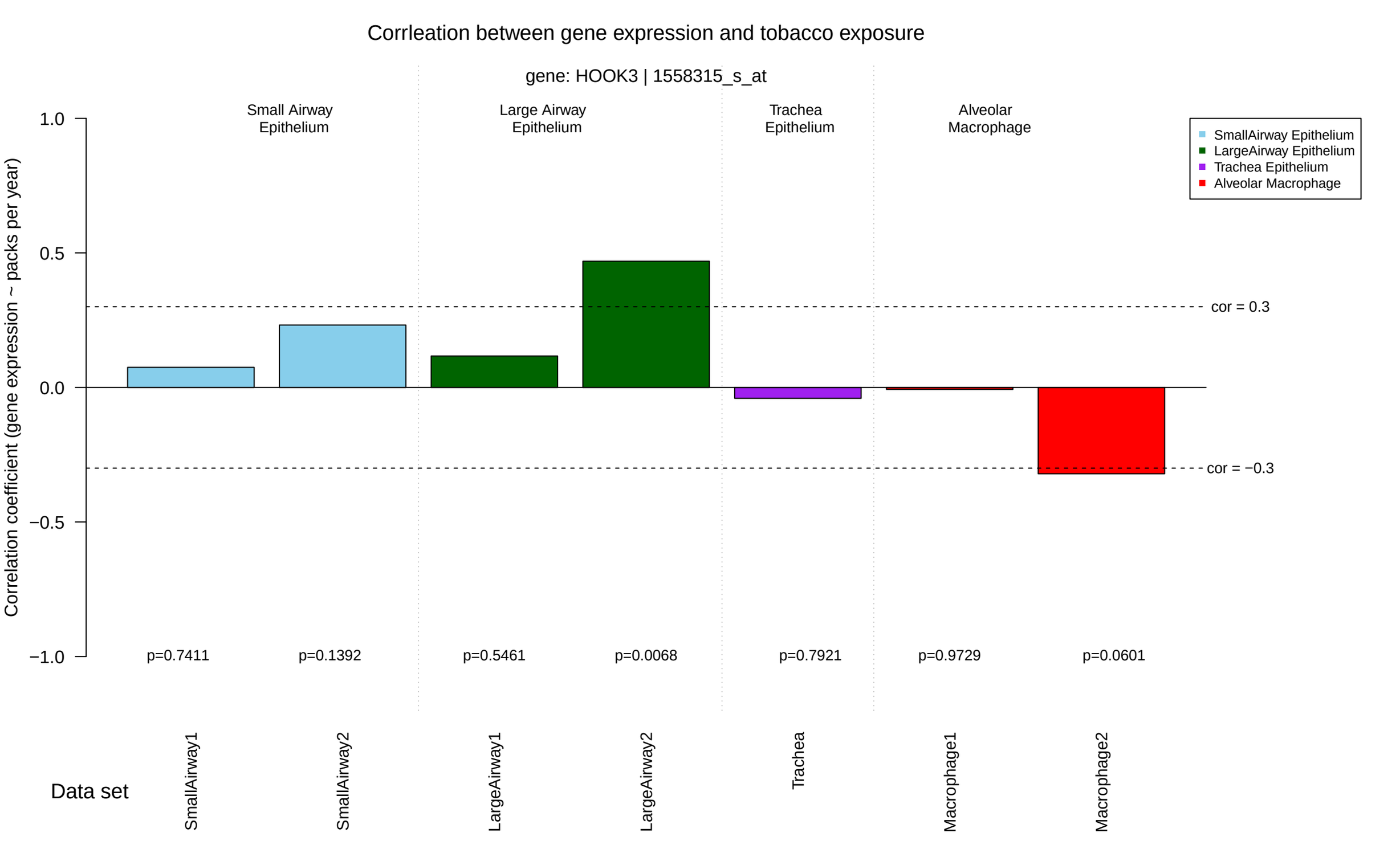

set: LargeAirway1 (p=0.37)
HOOK3 | 1558315_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4168)
HOOK3 | 1558315_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5961)
HOOK3 | 1558315_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3038)
HOOK3 | 1558315_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2066)
HOOK3 | 1558315_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOOK3 | 1558315_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7411
p=0.1392
p=0.5461
p=0.0068
p=0.7921
p=0.9729
p=0.0601
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2